

Eclipse 'Oxygen' debuts with Java 9 support

Eclipse Foundation has announced its annual release train called 'Oxygen' that upgrades a total of 83 projects. Alongside many new tweaks, the latest release highlights Java 9 support as the most anticipated upgrade.

The simultaneous releases by Eclipse Foundation are the result of coordinated efforts and communication across all Eclipse Open Source Project teams. The new release train includes two million new lines of code, and comes with improved functionality and performance across all the projects.

Java 9 support is still at the beta stage as the final build of JDK 9 is not out yet. However, the latest integration will help users add Java Runtime Environment for Java 9 as well as backing for the Java 9 execution environment. Developers can also leverage the Oxygen release for compiling modules using Java 9.

"The Eclipse Oxygen release includes many improvements in functionality and performance, includes new tools for Java code coverage analysis, and can be extended to support Java 9 development via an early access preview," the Eclipse team writes in an official announcement.

In addition to Java 9, the latest Eclipse Linux Tools 6.0 has enhanced the security of Docker tools. Developers can now also use C/C++ IDEs with Eclipse Linux Tools 6.0. Similarly, Eclipse PDT (PHP Development Tools) has added support for PHP v7.1.

The Oxygen release also updates the Eclipse Sirius 5.0 platform. You can use the platform for building domain-specific modelling tools. Additionally, the Foundation has worked on enhancing the performance and usability of Java.

are designed to help developers improve the security and performance of application containers.

The open source Smith project is designed to help developers build secure application containers. Oracle had to change the container build process to be comfortable with using containers in production. Therefore, the company developed a method to run containers while improving the stability and security of their environment.



"We have run into a number of operational issues with conventional container build processes. Smith is a tool that solves these issues by making

container builds more consistent and secure," said Vish Abrams, architect of cloud development, Amazon.

In addition to the Smith project, Railcar has been developed by Oracle as an alternative to existing container runtimes including *runc* used in Docker containers, which is written using the Rust programming language.

Rust offers performance and security benefits like never before. Also, Rust support enables Railcar to be compliant with the Open Container Initiative (OCI) runtime specification.

Abrams believes Rust is sitting at the intersection of C and Go languages — offering memory safety and higher-level primitives without sacrificing low-level control over threading. "It is a great choice for container utilities, and we hope to see the Rust community and the container community collaborate more in the future," he added.

The last tool launched by Oracle is Crashcart. The debugging tool is designed to improve the performance of container applications. It is capable of identifying the operational issues with the application container and suggesting the fix. Besides, the tool can access the file system of the container to identify issues.

All three open source launches are likely to give an edge to Oracle's OpenStack platform over time.

Rockstor Linux now comes with a feature to fully encrypt your disk

Rockstor, the open source Linux and Btrfs-based open source storage operating system, has received an update to version 3.9.1. This advanced version has plenty of new capabilities.



The major highlight of the update is full disk encryption using LUKS. The development team has also added support for scheduling power management and jumbo frames. There is a 4.10 kernel from ELRepo and a btrfs-progs update. A known systemd

issue has been fixed in the Rockstor 3.9.1 release. Also, the platform now shuts down cron jobs and imports an unlabelled Btrfs RAID5 pool.

Notably, the dated Linux kernel branch reached the end of life in May 2017. Users will, thus, no longer receive the security patches for the kernel.

You can download the Rockstor Linux 3.9.1 release from its official website. The update is a joint effort of six contributors, who addressed more than 30 issues in the previous Rockstor Linux release.